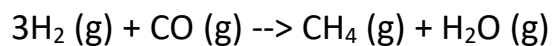


E16 Rate determining step

This is example 5.9 from Rawlings Ekerdt.

The synthesis of methane is carried out over a ruthenium catalyst:



One possible mechanism is

1. $\text{CO} (\text{g}) + \text{S} \rightleftharpoons \text{CO}_{\text{ads}}$
2. $\text{CO}_{\text{ads}} + \text{S} \rightleftharpoons \text{C}_{\text{ads}} + \text{O}_{\text{ads}}$
3. $\text{O}_{\text{ads}} + \text{H}_2 (\text{g}) \rightarrow \text{H}_2\text{O} (\text{g}) + \text{S}$
4. $\text{H}_2 (\text{g}) + 2\text{S} \rightleftharpoons 2\text{H}_{\text{ads}}$
5. $\text{C}_{\text{ads}} + \text{H}_{\text{ads}} \rightleftharpoons \text{CH}_{\text{ads}} + \text{S}$
6. $\text{CH}_{\text{ads}} + \text{H}_{\text{ads}} \rightleftharpoons \text{CH}_{2,\text{ads}} + \text{S}$
7. $\text{CH}_{2,\text{ads}} + \text{H}_{\text{ads}} \rightleftharpoons \text{CH}_{3,\text{ads}} + \text{S}$
8. $\text{CH}_{3,\text{ads}} + \text{H}_{\text{ads}} \rightarrow \text{CH}_4 (\text{g}) + 2\text{S}$

Step 8 is the RDS. Write the rate law for this reaction in terms of only reactant and product species, i.e., do not include any surface concentrations of intermediate species.

Decision Guide: Rate Determining Step (RDS) Example

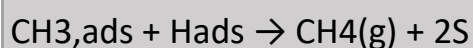
Goal: Derive a rate law that contains only: CO, H₂, CH₄, H₂O

and does **not** contain any surface intermediates such as: Coads, Cads, Oads, Hads, Chads, CH_{2,ads}, CH_{3,ads}

The problem states that **Step 8 is the rate-determining step (RDS).**

Step 1: Identify the Rate-Determining Step

Locate the step identified as the RDS:



This is Step 8. Because it is the slow step:

Overall Rate $\approx$ Rate of Step 8

Step 2: Write the Initial Rate Expression

The rate law comes directly from the elementary RDS: $r = k_8 C_{\text{CH}_3,\text{ads}} C_{\text{H},\text{ads}}$

or equivalently, Rate $\propto$ CH_{3,ads} $\times$ Hads

At this point the rate law is not yet acceptable because it contains surface intermediates.

Step 3: Identify Which Steps Are Assumed to Be at Equilibrium

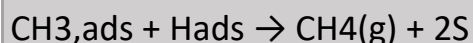
Because Step 8 is the only slow step, the preceding steps are assumed to be quasi-equilibrated:

Step 1, Step 2, Step 4, Step 5, Step 6, Step 7

These equilibrium relationships will be used to eliminate the surface intermediates.

Step 4: Work Backward Through the Mechanism

Starting from the RDS:



the initial rate expression is: $\text{Rate} \propto \text{CH}_3,\text{ads} \times \text{Hads}$

However, the final rate law cannot contain surface intermediates. Therefore, begin replacing the surface species using the equilibrium relationships from the fast steps. Work backward through the hydrogenation sequence:

CH_3,ads is related to CH_2,ads through Step 7. Then:

CH_2,ads is related to Chads through Step 6. Then:

Chads is related to Cads through Step 5.

Continue substituting until the rate expression is written in terms of: Cads , Hads , Vacant Sites (S) only.

Why Work Backward?

The rate-determining step contains:

CH_3,ads and Hads

but the problem requires a rate law containing only measurable reactant and product species. Therefore:

CH_3,ads must be replaced using Step 7, then CH_2,ads must be replaced using Step 6, then Chads must be replaced using Step 5, and so on until all adsorbed intermediates have been eliminated.

Concept Check

Write RDS Rate Expression

↓

Contains CH_3,ads and Hads

↓

Use Step 7

↓

Eliminate CH_3,ads

↓

Use Step 6

↓

Eliminate CH_2, ads

↓

Use Step 5

↓

Eliminate CHads

↓

Continue Until Only Gas-Phase Species Remain

The key idea is that every adsorbed intermediate appearing in the RDS rate law must eventually be replaced using the equilibrium relationships from the fast steps.

Step 5: Replace Surface Carbon Species

Use Step 2: $\text{COads} + \text{S} \rightleftharpoons \text{Cads} + \text{Oads}$

and Step 1: $\text{CO} + \text{S} \rightleftharpoons \text{COads}$

to express: Cads in terms of CO , Oads , and S

This removes the carbon-containing surface intermediates from the rate law.

Why Are We Doing This?

At the end of Step 4, the rate expression still contains adsorbed carbon species such as Cads . However, the problem requires a rate law written only in terms of measurable reactant and product species. Because Steps 1 and 2 are assumed to be at equilibrium, they provide relationships that connect Cads to CO , Oads , and Vacant Sites (S). Using these equilibrium expressions allows us to eliminate the carbon-containing surface intermediates from the rate law.

Concept Check

CO

↓

COads

↓

Cads

The equilibrium steps allow us to work backward from adsorbed carbon species to gas-phase CO. The goal is to continue replacing adsorbed intermediates until only observable gas-phase species remain in the final rate law.

Workflow

Cads

↓

Use Step 2 Equilibrium

↓

COads

↓

Use Step 1 Equilibrium

↓

CO

Each substitution removes one surface intermediate and moves the rate expression closer to the required final form.

Step 6: Replace Surface Hydrogen Species

Use Step 4: $\text{H}_2(\text{g}) + 2\text{S} \rightleftharpoons 2\text{Hads}$

to express Hads in terms of H₂ and S. This removes the hydrogen-containing surface intermediates from the rate law.

Why Are We Doing This?

At this stage, the rate expression still contains Hads. However, the final rate law must be written in terms of measurable gas-phase species. Because Step 4 is assumed to be at equilibrium, it provides a relationship between Hads, H₂, and Vacant Sites (S). Using this equilibrium expression allows us to replace the adsorbed hydrogen species with gas-phase hydrogen.

Concept Check

H₂

↓

Hads

The equilibrium adsorption step allows us to connect surface hydrogen species to the measurable hydrogen concentration or partial pressure.

Workflow

Hads

↓

Use Step 4 Equilibrium

↓

H₂ and S

After this substitution, the rate law contains fewer surface intermediates and moves closer to the desired form involving only reactants and products.

Step 7: Eliminate Oads

Now the rate expression still contains Oads

Use Step 3: $\text{Oads} + \text{H}_2(\text{g}) \rightarrow \text{H}_2\text{O}(\text{g}) + \text{S}$

to relate Oads to H₂, H₂O, and S. This removes the oxygen-containing surface intermediate from the rate law.

Why Are We Doing This?

At this point, most of the carbon-containing and hydrogen-containing surface intermediates have been eliminated.

However, the rate expression still contains Oads which is not an observable gas-phase species. Because Step 3 connects adsorbed oxygen to gas-phase species, it allows us to replace Oads with a relationship involving H₂, H₂O, and Vacant Sites (S). This moves the rate law closer to the desired form containing only reactants and products.

Workflow

Oads

↓

Use Step 3 Relationship



H₂ and H₂O



Eliminate Surface Oxygen Species

After this step, all surface species have been connected to measurable gas-phase species.

Step 8: Apply the Site Balance

A catalyst surface contains a finite number of sites.

Write the site balance:

$$C_T = C_S + C_{CO,ads} + C_{C,ads} + C_{O,ads} + C_{H,ads} + C_{CH,ads} + C_{CH_2,ads} + C_{CH_3,ads}$$

where:

C_T = total concentration of catalytic sites, C_S = concentration of vacant sites, and the remaining terms represent occupied sites. Use the equilibrium relationships derived above to solve for C_S (vacant-site concentration).

Step 9: Substitute into the Rate Law

Substitute: CH_{3,ads}, H_{ads}, and S with the expressions derived above into the rate law.

Step 10: Simplify

Collect all equilibrium constants and kinetic constants into an overall rate constant. The final expression should resemble a typical Langmuir-Hinshelwood form:

Rate = (Numerator involving reactant partial pressures) / (Adsorption term involving surface coverage)

Final Answer Checklist

- ✓ Identified the rate-determining step.
- ✓ Wrote the elementary rate law for the RDS.
- ✓ Used equilibrium assumptions for the fast steps.
- ✓ Eliminated CH_3, ads .
- ✓ Eliminated Hads .
- ✓ Eliminated Cads and Oads .
- ✓ Applied the site balance.
- ✓ Obtained a rate law containing only reactant and product species.

Key Insight

Write Rate of RDS

↓

Use Fast-Step Equilibria

↓

Replace Surface Species

↓

Apply Site Balance

↓

Obtain Observable Rate Law

For rate-determining-step problems, the hardest part is usually **not writing the RDS rate expression**. The difficult part is systematically eliminating every surface intermediate until the final rate law depends only on measurable reactant and product concentrations or partial pressures.